

AFRILANG Stdlib

std/c/mlkmeans

Français. Bibliothèque AFRILANG 'std/c/mlkmeans' (niveau complex, catégorie complex). Elle expose 6 fonction(s) : kmeans, assignClusters, inertia, clusterSizes, silhouetteSample, elbowK.

'import "std/c/mlkmeans"' · 'use mlkmeans'

- 'kmeans(points list number, k number, dims number) → list number' - 'assignClusters(points list number, centroids list number, dims number) → list number' - 'inertia(points list number, centroids list number, dims number) → number' - 'clusterSizes(labels list number, k number) → list number' - 'silhouetteSample(points list number, labels list number, idx number, dims number

English. AFRILANG library 'std/c/mlkmeans' (tier complex, category complex). Exports 6 function(s): kmeans, assignClusters, inertia, clusterSizes, silhouetteSample, elbowK.

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Catégorie / Category	Modules complex (c/)	
Niveau / Tier	complex	June 16, 2026
Fonctions / Functions	6	

Contents

Français

1. Vue d'ensemble

Bibliothèque AFRILANG 'std/c/mlkmeans' (niveau complex, catégorie complex). Elle expose 6 fonction(s) : kmeans, assignClusters, inertia, clusterSizes, silhouetteSample, elbowK.

```
'import "std/c/mlkmeans"' · 'use mlkmeans'
```

```
- `kmeans(points list number, k number, dims number) → list number`
- `assignClusters(points list number, centroids list number, dims number) → list number`
- `inertia(points list number, centroids list number, dims number) → number`
- `clusterSizes(labels list number, k number) → list number`
- `silhouetteSample(points list number, labels list number, idx number, dims number) → number`
- `elbowK(points list number, maxK number, dims number) → number`
```

2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/mlkmeans"
use mlkmeans
```

Niveau : complex · Catégorie : Modules complex (c/)
Domaines avancés – graphes, NLP, réseaux complexes.

3. Référence API

- kmeans(points list number, k number, dims number) → list•
- assignClusters(points list number, centroids list number, dims number) → list•
- inertia(points list number, centroids list number, dims number) → number•
- clusterSizes(labels list number, k number) → list•
- silhouetteSample(points list number, labels list number, idx number, dims number) → number•
- elbowK(points list number, maxK number, dims number) → number

4. Exemples d'utilisation

```
import "std/c/mlkmeans"
use mlkmeans
```

```
create result = kmeans(/* args */)
say result
```

```
create result = assignClusters(/* args */)
say result
```

```
create result = inertia(/* args */)
say result
```

```
create result = clusterSizes(/* args */)
say result
```

```
create result = silhouetteSample(/* args */)
say result
```

```
create result = elbowK(/* args */)
say result
```

5. Bonnes pratiques

- Préférez ``import "std/c/mlkmeans"`` en tête de fichier.
- Lancez ``afriLang check`` avant de livrer.
- Combinez avec les modules `stdlib` de la même catégorie.
- Voir `/stdlib/` pour le source live et les démos playground.
- Signalez les problèmes sur GitHub si une export manque.

6. Annexe — source

module `mlkmeans`

```
function kmeans(points list number, k number, dims number) returns list number
end
```

```
function assignClusters(points list number, centroids list number, dims number)
  returns list number
end
```

```
function inertia(points list number, centroids list number, dims number) returns
  number
end
```

```
function clusterSizes(labels list number, k number) returns list number
end
```

```
function silhouetteSample(points list number, labels list number, idx number, dims
  number) returns number
end
```

```
function elbowK(points list number, maxK number, dims number) returns number
end
end
```

English

1. Overview

AFRILANG library 'std/c/mlkmeans' (tier complex, category complex). Exports 6 function(s): kmeans, assignClusters, inertia, clusterSizes, silhouetteSample, elbowK.

'import "std/c/mlkmeans"' · 'use mlkmeans'

```
- `kmeans(points list number, k number, dims number) → list number`
- `assignClusters(points list number, centroids list number, dims number) → list number`
- `inertia(points list number, centroids list number, dims number) → number`
- `clusterSizes(labels list number, k number) → list number`
- `silhouetteSample(points list number, labels list number, idx number, dims number) → number`
- `elbowK(points list number, maxK number, dims number) → number`
```

2. Installation & import

Add the module to your program:

```
import "std/c/mlkmeans"
use mlkmeans
```

Tier: complex · Category: Complex modules (c/)
Advanced domains – graphs, NLP, complex networks.

3. API reference

- kmeans(points list number, k number, dims number) → list•
- assignClusters(points list number, centroids list number, dims number) → list•
- inertia(points list number, centroids list number, dims number) → number•
- clusterSizes(labels list number, k number) → list•
- silhouetteSample(points list number, labels list number, idx number, dims number) → number•
- elbowK(points list number, maxK number, dims number) → number

4. Usage examples

```
import "std/c/mlkmeans"
use mlkmeans
```

```
create result = kmeans(/* args */)
say result
```

```
create result = assignClusters(/* args */)
say result
```

```
create result = inertia(/* args */)
say result
```

```
create result = clusterSizes(/* args */)
say result
```

```
create result = silhouetteSample(/* args */)
say result
```

```
create result = elbowK(/* args */)
say result
```

5. Best practices

- Prefer ``import "std/c/mlkmeans"`` at the top of your file.
- Run ``afriLang check`` before shipping.
- Combine with related stdlib modules from the same category.
- See `/stdlib/` for the live source and playground demos.
- Report issues on GitHub if an export is missing or undocumented.

6. Appendix — source

module mlkmeans

```
function kmeans(points list number, k number, dims number) returns list number
end
```

```
function assignClusters(points list number, centroids list number, dims number)
  returns list number
end
```

```
function inertia(points list number, centroids list number, dims number) returns
  number
end
```

```
function clusterSizes(labels list number, k number) returns list number
end
```

```
function silhouetteSample(points list number, labels list number, idx number, dims
  number) returns number
end
```

```
function elbowK(points list number, maxK number, dims number) returns number
end
end
```

Référence rapide / Quick reference

- Import : `import "std/c/mlkmeans"`
- Vérification : `afriLang check`
- Documentation : <https://afriLang.dev/stdlib/std/c/mlkmeans/>

Source (extrait)

```
module mlkmeans

function kmeans(points list number, k number, dims number) returns list number
end

function assignClusters(points list number, centroids list number, dims number)
  returns list number
```

```
end

function inertia(points list number, centroids list number, dims number) returns
    number
end

function clusterSizes(labels list number, k number) returns list number
end

function silhouetteSample(points list number, labels list number, idx number, dims
    number) returns number
end

function elbowK(points list number, maxK number, dims number) returns number
end
end
```